$\therefore\ 2 \le \dim(U+W) \le 4$.

05.02.2020

Nullity of T :-

Let $T: U \to V$ be a linear map over the same field, then null space (Kernel) of T is defined as,

$$N(T) = \{u \in U : T(u) = 0\} = \text{Ker}(T)$$

$$R(T) = \{v \in V : T(u) = v\}$$

* $N(T)$ is a subspace of U.

Let $u_1, u_2 \in N(T)$

$\Rightarrow u_1, u_2 \in U$ s.t. $T(u_1) = 0$, $T(u_2) = 0$

$\Rightarrow \alpha u_1 + \beta u_2 \in U$

$$T(\alpha u_1 + \beta u_2) = \alpha T(u_1) + \beta T(u_2) \quad [\because T \text{ is linear}]$$
$$= \alpha \cdot 0 + \beta \cdot 0$$
$$= 0$$

$\therefore \alpha u_1 + \beta u_2 \in N(T)$.

$\therefore N(T)$ is a subspace of U.

* $R(T)$ is a subspace of V.

Proof:- Let $v_1, v_2 \in R(T)$

$\Rightarrow v_1, v_2 \in V$ s.t. $T(u_1) = v_1$, $T(u_2) = v_2$

$\alpha v_1 + \beta v_2 \in V \quad [\because V \text{ is a v.s}]$

Now, $u_1, u_2 \in U$

$\Rightarrow \alpha u_1 + \beta u_2 \in U$

$$T(\alpha u_1 + \beta u_2) = \alpha T(u_1) + \beta T(u_2) \quad [\because T \text{ is linear}]$$
$$= \alpha v_1 + \beta v_2$$

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